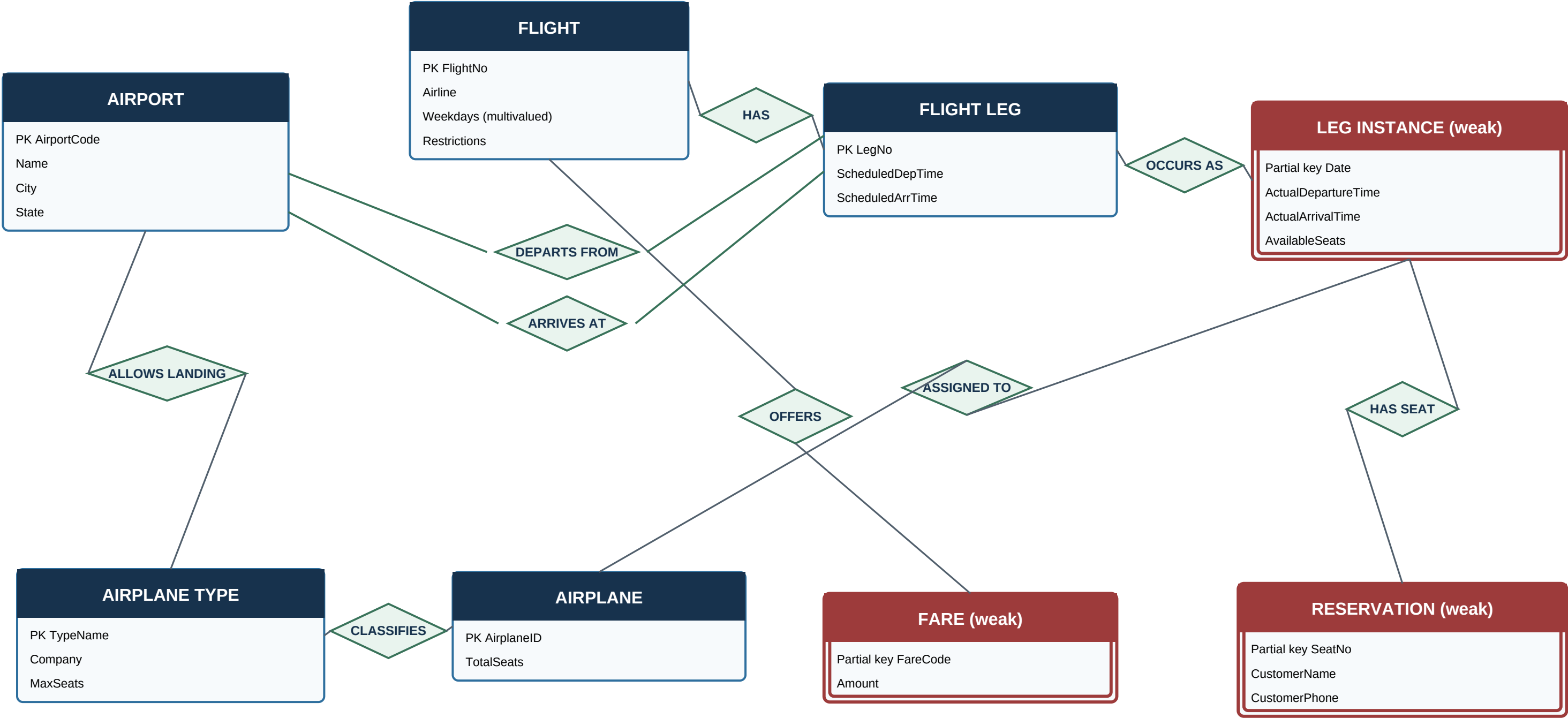


ERD Task 1 - Airline Information System

Entities, attributes, cardinality and participation

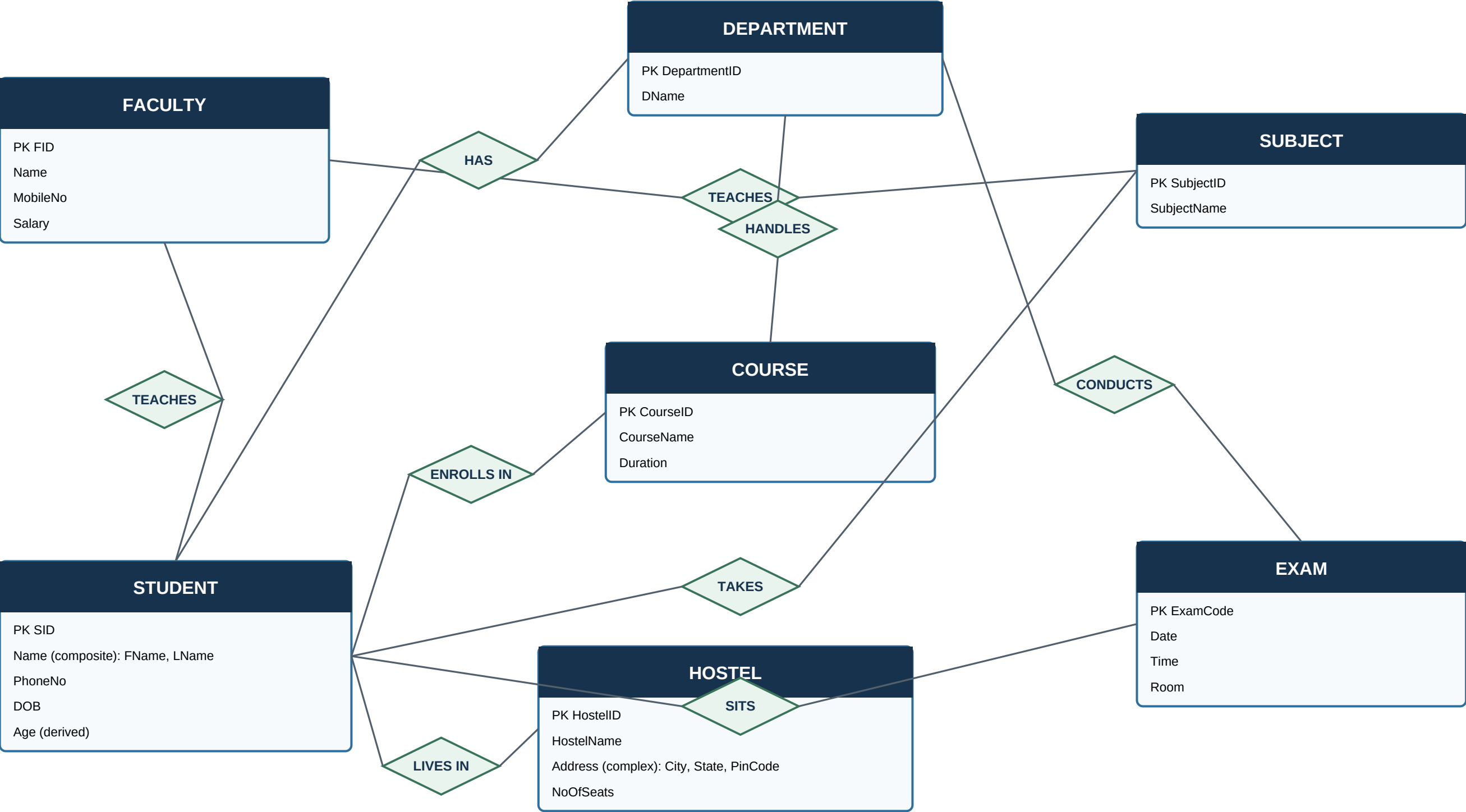


Notation and relationship rules

PK = primary key | AK = alternate/candidate key | Partial key identifies a weak entity with its owner | 0 = optional | 1 = mandatory
FLIGHT 1 - N FLIGHT LEG | FLIGHT LEG 1 - N LEG INSTANCE | each leg has exactly 1 departure airport and 1 arrival airport
AIRPLANE TYPE 1 - N AIRPLANE | AIRPLANE 1 - N LEG INSTANCE | AIRPORT M - N AIRPLANE TYPE | LEG INSTANCE 1 - N RESERVATION
Keys: LEG INSTANCE = LegNo + Date | RESERVATION = LegNo + Date + SeatNo | FARE = FlightNo + FareCode

ERD Task 2 - College Management System

Entities, attributes, cardinality and participation

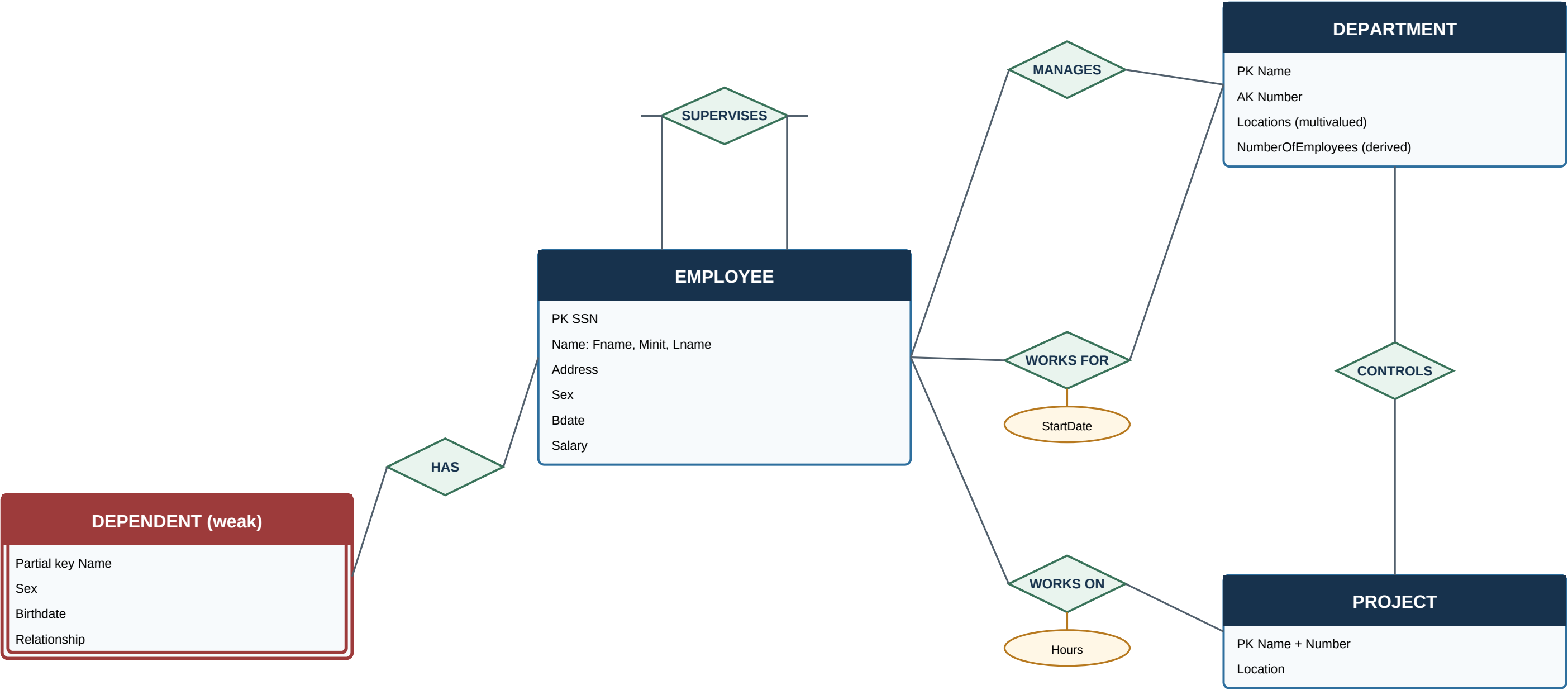


Notation and relationship rules

PK = primary key | AK = alternate/candidate key | Partial key identifies a weak entity with its owner | 0 = optional | 1 = mandatory
DEPARTMENT 1 - N COURSE | DEPARTMENT 1 - N EXAM | DEPARTMENT 1 - N STUDENT | HOSTEL 1 - N STUDENT (student participation optional)
STUDENT M - N COURSE | STUDENT M - N SUBJECT | STUDENT M - N EXAM | FACULTY M - N SUBJECT | FACULTY M - N STUDENT
Age is derived from DOB | Student Name is composite | Hostel Address is complex | M:N relationships become associative tables during mapping

ERD Task 3 - Company Database

Entities, attributes, cardinality and participation



Notation and relationship rules

PK = primary key | AK = alternate/candidate key | Partial key identifies a weak entity with its owner | 0 = optional | 1 = mandatory
WORKS FOR: EMPLOYEE 1 total - DEPARTMENT 0..N partial | MANAGES: EMPLOYEE 0..1 partial - DEPARTMENT 1 total
CONTROLS: DEPARTMENT 0..N partial - PROJECT 1 total | WORKS ON: EMPLOYEE 0..N partial - PROJECT 0..N partial
DEPENDENT is weak | Hours belongs to WORKS ON | StartDate belongs to WORKS FOR | Locations multivalued | NumberOfEmployees derived